Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Kushmerick, Martin J M.D., Ph.D.		
Project Title:	Integrating Human Muscle Energetics and Mechanics		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98195	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	02/01/2001	**End Date:**	05/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	IHF.002.02 MA.002.12		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Statement of Research Aims: Aim 1: Measure the economy and efficiency of human muscle contraction and sustainable power output. Hypotheses tested: 1. The balance between ATP supply and ATP demands account quantitatively for the difference in sustained performance (duty cycle) in various muscles and individuals. 2. Working contractions add a substantially larger, myofibrillar cost above isometric twitches in which only ion transport activation costs dominate. 3. Muscles differ in economy and efficiency: mechanically slower muscles have higher economy and thermodynamic efficiency for converting ATP to external work than fast muscles, but lower power. Aim 2: Integrate the component mechanisms of energy supply and demand into models to make predictions of energy costs and balance in new experimental conditions. Hypotheses tested: 1. Simple models of the components of energy balance as developed in aim 1 are necessary and sufficient to account for the major energetic properties of human muscle. 2. The models establish a cellular basis for defining isometric economy and working efficiency. 3. Variations in		

	properties among muscles and individuals define the normal distribution of properties so this distribution can be used to define probabilistic responses of the system. 4. When new mechanistic components at the molecular and cellular level are added, they can be tested for their effects on the system operation. The results expected would show that the diversity of muscles is quantitatively explained by these elemental components and the principles of energy balance in a parsimonious way. This is a major integration of mechanisms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall goal of this project is to build up a series of measurements and models of intracellular energy and metabolic fluxes (ATP supply and demand), mechanics during exercise (ATP demand), and blood flow (ATP supply) to make possible an integration of the mechanisms involved and a validation of models describing these processes. The strategy for integrating this project with other parts of the Integrated Human Function team and with other NSBRI teams is to work this process out for one limb muscle, as a paradigm for other project integrations, and then identify the differences in skeletal muscle properties among individuals within the context of limb function. For the NASA critical path analyses, this work will establish a basis for interpreting the fundamental properties of muscle. On board the ISS and other space vehicles, ultrasound will be available to make measurements similar to those conducted in this project. This project will provide criteria for judging the effectiveness of countermeasures on human limb muscle function and tools for monitoring limb functional properties in space.
Task Progress:	No progress report this period. See bibliographic publication.
Bibliography Type:	Description: (Last Updated: 09/12/2017)
Articles in Peer-reviewed Journals	Jubrias SA, Vollestad NK, Gronka RK, Kushmerick MJ. "Contraction-coupling efficiency of human first dorsal interosseous muscle." J Physiol. 2008 Apr 1;586(7):1993-2002. Epub 2008 Jan 31. https://doi.org/10.1113/jphysiol.2007.146829 ; PubMed PMID: 18238810; PubMed Central PMCID: PMC2375725 , Apr-2008